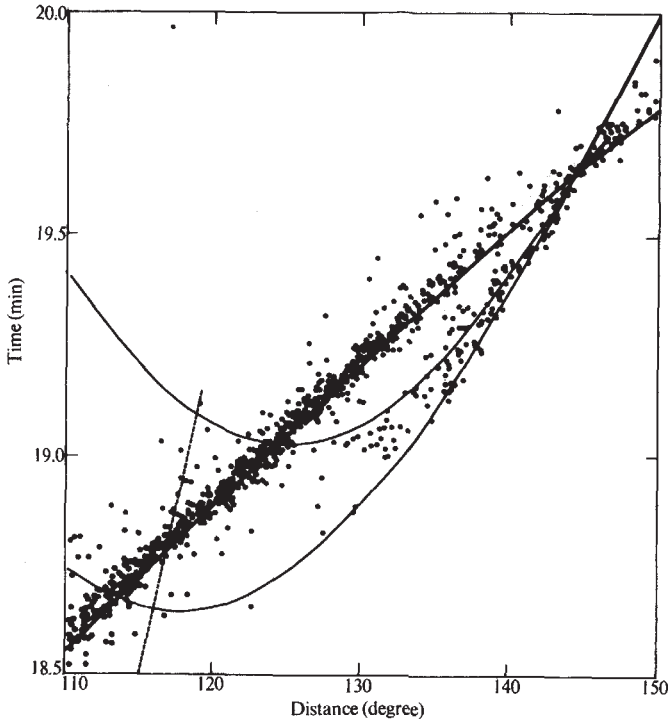


Fig. 3 Data listed as PKIKP residuals in ISC bulletins (Table 1). The solid curves are the same as in Fig. 1. The dashed line is an extension of the PdP curve of Bolt *et al.*[18].

PKKP are to be expected on the present interpretation, but may be difficult to detect in the codas of preceding signals.

In conclusion, we note that the inferred existence of transition layers in the outer core is presently based entirely on the evidence of precursors to PKP. The re-interpretation of precursors presented here removes the need to postulate structural complexities in this region.

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Jurassic Palaeomagnetism

WE have presented some tentative palaeomagnetic results from a Jurassic sedimentary sequence in southwestern USA[1]. The results suggested the existence of a normal and reversed polarity sequence in rocks whose stability had been tested by thermal demagnetization. We have subsequently carried out a sedimentological study of these rocks, the results of which have changed our original conclusions.

Although haematite was found at nearly all grain contacts indicating that it was introduced early in the diagenetic history of these rocks, it is also found both as a staining on the secondary quartz overgrowths and in pore spaces. It is this latter secondarily formed haematite which forms the bulk of the haematite present and, we now believe, is primarily responsible for the remanence we measured.

This discovery led us to re-examine our previous results. The results from the Navajo Sandstone which provided differing results from the upper and lower parts of the section were from rocks which had been collected from two different parts of the Gunlock section (S. Utah) each with a characteristic dip. We have recomputed all the measurements with respect to the present horizontal, the closest approach to a fold test possible, and found that the difference in direction between the two sections was minimized with the virtual pole (8.72 N, 26.0 E, $\alpha = 3.1$) not significantly different from the present geographic pole. We therefore feel that other Jurassic formations which provide similar virtual pole positions to those we originally calculated from the bedding plane position may similarly be suspect (Table 1).

We believe the magnetization of the immediately underlying Kayenta Formation is stable and represents the geomagnetic field direction of Upper Triassic times, for not only did we find the magnetization stable, but the directions are consistent with those of more fully investigated rocks of the same age

Table 1 Jurassic Virtual Pole Positions for North America

Formation	Sampling*	Decl.	Inc.	$\alpha 95$	K	Virtual pole Long.	Virtual pole Lat.	Reference
Navajo Sandstone lower corrected to Jur. hor.	107S, 480 sp.	347.8	35.6	5.5	78.1	72.1 E	80.0 N	4
Navajo Sandstone upper corrected to Jur. hor.	69S, 280 sp.	13.1	49.4	3.5	196.9	5 W	81.8 N	4
Navajo Sandstone corrected to present hor.	176S, 760 sp.	1.8	54.3	3.1	113.5	36.0 E	87.2 N	4
Navajo Sandstone reversed	20 sp.	162.2	9.6	40.5	1.6	94.1 E	43.0 N	4
Kayenta Fm.	13S, 104 sp.	351.0	19.0	9.8	50.9	82.5 E	61.2 N	4
Kayenta Fm.	4S, 39 sp.	16.0	50.0	—	—	39.0 E	83.0 N	5
Carmel Fm.	9S, 31 sp.	349.0	63.0	14.0	11.0	160 W	80.0 N	5
Carmel Fm. correct to Jur. hor.	12S, 48 sp.	18.9	64.2	11.5	13.4	52.8 W	75.5 N	4
Carmel Fm. correct to present hor.	12S, 48 sp.	352.6	73.0	11.1	13.6	122 W	70.1 N	4

*S = sites; sp. = samples.

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elsewhere. The discovery of the secondary magnetization of the Jurassic Navajo Sandstone removes the high rate of displacement which is necessary to explain the change between the virtual pole positions derived from the Kayenta and Navajo rocks.

The reversed directions of magnetization found in the Navajo Sandstone are dispersed; however, the virtual pole position calculated from them (274.1 E, 43 S, $\alpha = 40.5$) differs little from that of the Kayenta Formation. It therefore appears to us that in these reversely magnetized samples we may be observing relicts of an original magnetization overprinted by a variable but small stable secondary magnetic component (of CRM type). It is probable that a similar situation may exist in the normally magnetized rocks, but is more difficult to discern.

The question of the time of origin of the apparently dominant normal direction of secondarily acquired magnetization is important in view of the rapidity of polarity reversals in the last few million years. We suggest the magnetization was most probably acquired during the Brunhes normal epoch when, owing to the strong climatic fluctuations of the Ice Age, water movement through the porous Navajo Sandstone was at a peak. This clearly implies a mobility of iron greater than is normally regarded to be the case. It is also clear that without a sedimentary examination the magnetization of porous rocks must be treated with some caution, and for this reason we regard our measurement on the Carmel Formation as suspect.

It is unfortunately true that the behaviour of the Jurassic geomagnetic field in North America still remains largely unknown, with the best results to date those of Symons[2] and Opdyke and Wensink[3].

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Concentration of Suspended Particulate Matter in Surface Sea Water

A KNOWLEDGE of the kind and amount of suspended particulate matter in the World Ocean is important for the solution of a number of geological, geochemical and environmental problems. Varying estimates of total suspended particulate matter in sea water have been made and Lisitzin[1] and Gordeyev[2] report concentrations which, in some areas, are one or two orders of magnitude higher than those found by Ewing *et al.*[3,4].

We present here the results of a recent study in which particulate matter was collected from surface water samples (0–5 m) from the Atlantic and Indian Oceans and the China Sea during March and July 1972. Collections were made on board MV Cyclops by pumping sea water from a position by the ship's bows (which minimized contamination from the ship's sides) *via* plastic tubing and filtering through weighed 0.45 μm membrane filters. The filters were washed with distilled water until salt-free, dried at 60° C to constant weight and the weight of particulate matter determined. The ship's route is shown in Fig. 1, and the results for the various oceanic areas are listed in Table 1. Some of the more important aspects of the study are discussed below.

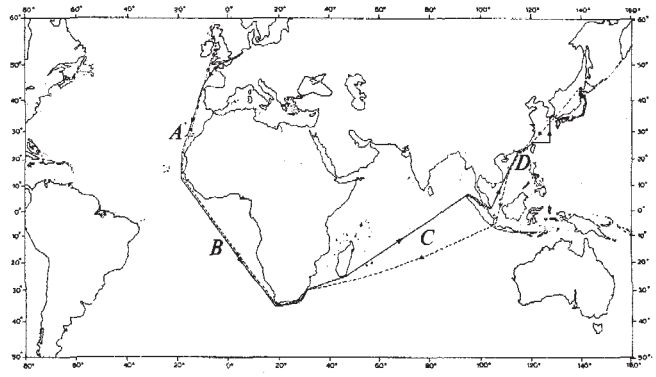


Fig. 1 The track of the MV Cyclops. ——, UK–Japan run; - - -, Japan–UK run. Average concentrations of total suspended matter in surface water are given for the major oceanic areas. *A*, 467 μg l.$^{-1}$; *B*, 150 μg l.$^{-1}$; *C*, 66 μg l.$^{-1}$; *D*, 127 μg l.$^{-1}$.

Table 1 Concentration of Total Suspended Particulate Matter in Surface Sea Waters

Oceanic area	No. of samples	Range of concentration (μg l.$^{-1}$)	Average concentration (μg l.$^{-1}$)	Time of year
North Atlantic	25	121–1,168	467	April and July
South Atlantic	23	67–448	150	April and July
Indian	30	43–110	66	April and June
China Sea	22	29–338	127	May and June
Oceanic deep-sea average (including N. Atlantic) *	100	29–1,168	199	—
Oceanic deep-sea average (excluding N. Atlantic)	75	29–448	110	—
Java Sea	2	580–915	747	June
English Channel	4	796–2,609	1,719	March
Irish Sea	6	1,087–1,960	1,680	July
Inland Sea (Japan)	7	300–3,641	1,452	May
South Japanese coast	7	173–349	245	May–June
South African coast	11	84–1,530	456	April and June

* Two oceanic deep-sea averages are given because collections in the North Atlantic included some from near-shore waters.

Along the eastern margins of the North Atlantic Ocean the average concentration of suspended matter during April and July was 467 μg l.$^{-1}$. Along a similar track Gordeyev[2] found an average of ~1,000 μg l.$^{-1}$ in April, and Svirenko[5] reported concentrations of 800–1,200 μg l.$^{-1}$; both sets of determinations were made with 0.7 μm filters. Our values are lower than these, but are higher than the 40–150 μg l.$^{-1}$ quoted for North Atlantic open-ocean surface water[7]. This is to be expected because our collections were made along the continental margins.

Our average value for the concentration of suspended matter in the open-ocean surface waters of the South Atlantic during April and July was 150 μg l.$^{-1}$. This is considerably lower than the average of ~800 μg l.$^{-1}$ found by Gordeyev[2] for surface waters in the same area in April, and approaches that of Jacobs and Ewing, who used a continuous centrifuge, and reported an average of ~50 μg l.$^{-1}$ for open-ocean South Atlantic waters below a depth of 180 m. In open-ocean water of the tropical Atlantic Svirenko gave an average of ~400 μg l.$^{-1}$ of suspended matter.

Jacobs and Ewing[3] found an average of ~71 μg l.$^{-1}$ in open-ocean Indian Ocean waters below 180 m. This is very close to our average of 66 μg l.$^{-1}$ for open-ocean surface waters. In contrast, Lisitzin reported concentrations of 500–1,000 μg l.$^{-1}$ in surface waters of the Indian Ocean.

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